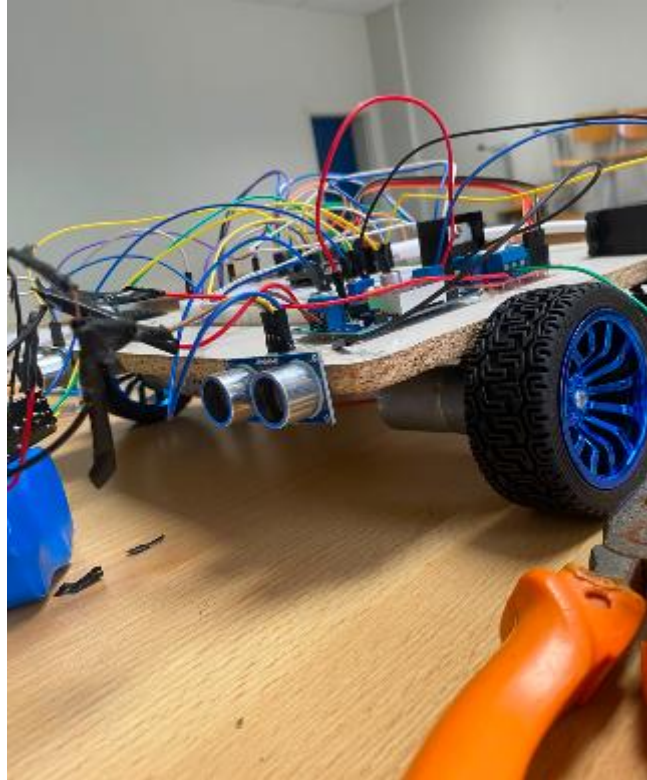


LEBANESE UNIVERSITY

Faculty of Engineering III
Department of Computer and Communications Engineering

AUTONOMOUS FIREFIGHTING ROBOT

Final Technical Report



Course	Sensors & Instrumentation
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Academic year	2025-2026

GitHub Repository: [Sensors-Instrumentation--Final-Project](#)

Demonstration Video: [Google Drive demo video](#)

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Abstract

This report presents the design, implementation, testing, and evaluation of an ESP32-based autonomous firefighting robot for indoor emergency-response scenarios. The prototype combines optical fire detection, temperature and humidity monitoring, water-flow measurement, line sensing, ultrasonic obstacle detection, motor control, a relay-driven water pump, a servo-mounted nozzle, an I2C LCD, and Bluetooth Low Energy communication. The robot can follow a simple line track, stop automatically when an obstacle is detected, detect a nearby flame, stop its drive motors and nozzle scan, and activate the pump. The YF-S201 flow sensor is used to calculate water flow and update the remaining tank level, while the LCD presents practical status information rather than the flow rate itself. The chassis was manufactured by the team from a 25 cm by 25 cm board, with manually prepared mounting holes for the motor and bearing assembly. Testing identified limitations in flame-sensor sensitivity, line-sensor stability, ultrasonic short-range behavior, water-system sealing, power delivery, and hardware reliability. A pressure sensor was removed after unsuccessful testing, and the load-cell/HX711 force subsystem was not integrated because the available ESP32 pins were already assigned to the main working subsystems. The final prototype demonstrated autonomous sensing and response, direct phone control for testing, live LCD visualization, and CSV-formatted serial logging.

Important compliance limitation

The project brief required a force-sensing modality. A load cell and HX711 were available, but they were not integrated into the final working prototype because of pin and integration constraints. This report documents that limitation rather than claiming a measurement that was not performed.

1. Introduction

Fire emergencies in indoor spaces require rapid detection and an immediate initial response. Entering a hazardous area may expose people to heat, smoke, and structural risks. A compact mobile sensing platform can reduce this risk by approaching the area, collecting environmental measurements, and performing a limited firefighting action before a person intervenes.

The objective of this project was to design and build an Autonomous Firefighting Robot controlled by an ESP32. The robot integrates environmental sensing, autonomous navigation, obstacle detection, water-flow monitoring, real-time display, wireless control, and actuator control in one prototype. The system was developed through iterative breadboard testing, mechanical assembly, individual module tests, integration, troubleshooting, and final demonstration.

The final robot can operate autonomously on a simple line track, stop in front of a close obstacle, detect a flame, stop its movement and servo motion, and activate a relay-controlled 12 V water pump. Bluetooth Low Energy is retained as an optional control and testing interface, while the automatic sensing functions operate without requiring phone commands.

2. Problem Statement

The engineering problem addressed by the project was how to create a low-cost mobile system that can sense an indoor fire-related condition, navigate a constrained path, avoid collision, and deliver water while providing the user with understandable real-time information.

The system was expected to:

- Detect a flame using an optical sensor.
- Measure ambient temperature and relative humidity.
- Measure water flow during pump operation.
- Follow a simple floor line without continuous phone control.

- Detect a nearby obstacle and stop before collision.
- Trigger an automatic response when fire is detected.
- Display the most important system information on an LCD.
- Log measurements in CSV format for later inspection.
- Allow direct Bluetooth control during testing and demonstration.

3. System Architecture

The ESP32 is the central controller. It samples the sensors, applies threshold and state logic, controls the motors, relay, pump, servo, and LCD, and communicates with a phone through BLE. Sensor acquisition and actuator control are implemented as repeated non-blocking update functions where possible, allowing several subsystems to operate in parallel.

Table 1. System architecture summary.

Subsystem	Main components	Function
Control	ESP32 development board	Reads inputs, executes decision logic, controls outputs, logs data, and manages BLE.
Sensing	Flame sensor, DHT11, YF-S201, line tracker, HC-SR04	Measures fire indication, temperature, humidity, water flow, path position, and obstacle distance.
Motion	L298N driver and four DC motors	Provides forward, backward, left, and right movement.
Fire response	Relay, 12 V pump, tubing, nozzle, MG996R servo	Scans the nozzle when idle and sprays water when fire is detected.
Interface	LCD1602 I2C and BLE phone application	Displays status and allows optional direct control/testing.
Power	12 V battery and XL4015 buck converter	Supplies 12 V loads and steps voltage down to approximately 5 V for suitable modules.

Autonomous Firefighting and Rescue Robot - Circuit / Wiring Diagram

ESP32 main controller with sensors, actuators, pump, flow sensor, LCD, and common ground

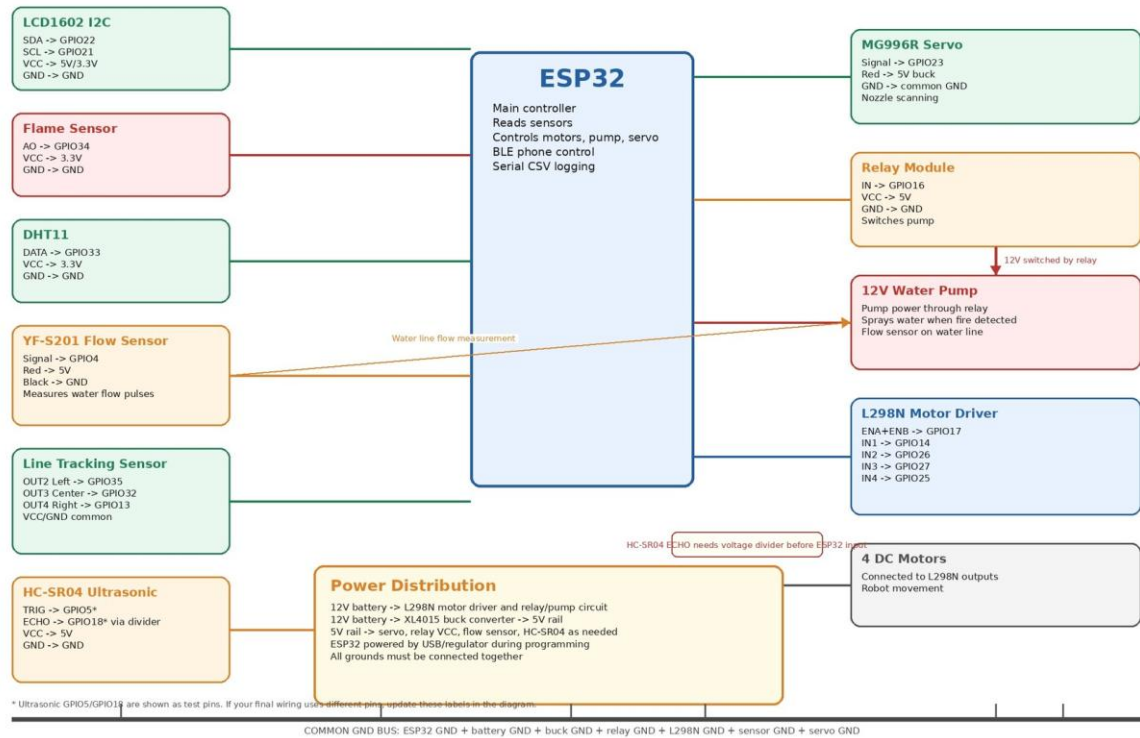


Figure 1. Circuit-level system diagram showing the ESP32, sensors, motor driver, relay, pump, servo, LCD, and power connections.

3.1 Operational States

The main operational behaviors are summarized below:

- **Normal monitoring:** the robot reads sensors, follows the line when autonomous movement is enabled, scans the nozzle while the pump is off, updates the LCD, and logs data.
- **Obstacle response:** when the measured ultrasonic distance reaches the short stopping threshold, the motors stop and the LCD displays an obstacle warning.
- **Fire response:** when the flame reading crosses the selected software threshold, the motors stop, the servo scan stops, and the pump is activated.
- **Low-water response:** the flow calculation reduces the stored tank value; the LCD presents a refill warning and the remaining amount or percentage.
- **Phone-control response:** movement, pump, and servo commands can be sent over BLE for testing or direct intervention.



Figure 2. Integrated robot prototype positioned on the line track during system testing.

4. Sensor Selection and Justification

4.1 Flame Sensor

A four-pin infrared flame-detection module was selected for optical fire detection. The digital output was initially tested, but its sensitivity was insufficient even after adjustment of the module potentiometer. The final implementation therefore used the analog output connected to GPIO34. This allowed the raw ADC reading to be observed in the Serial Monitor and a software threshold to be selected.

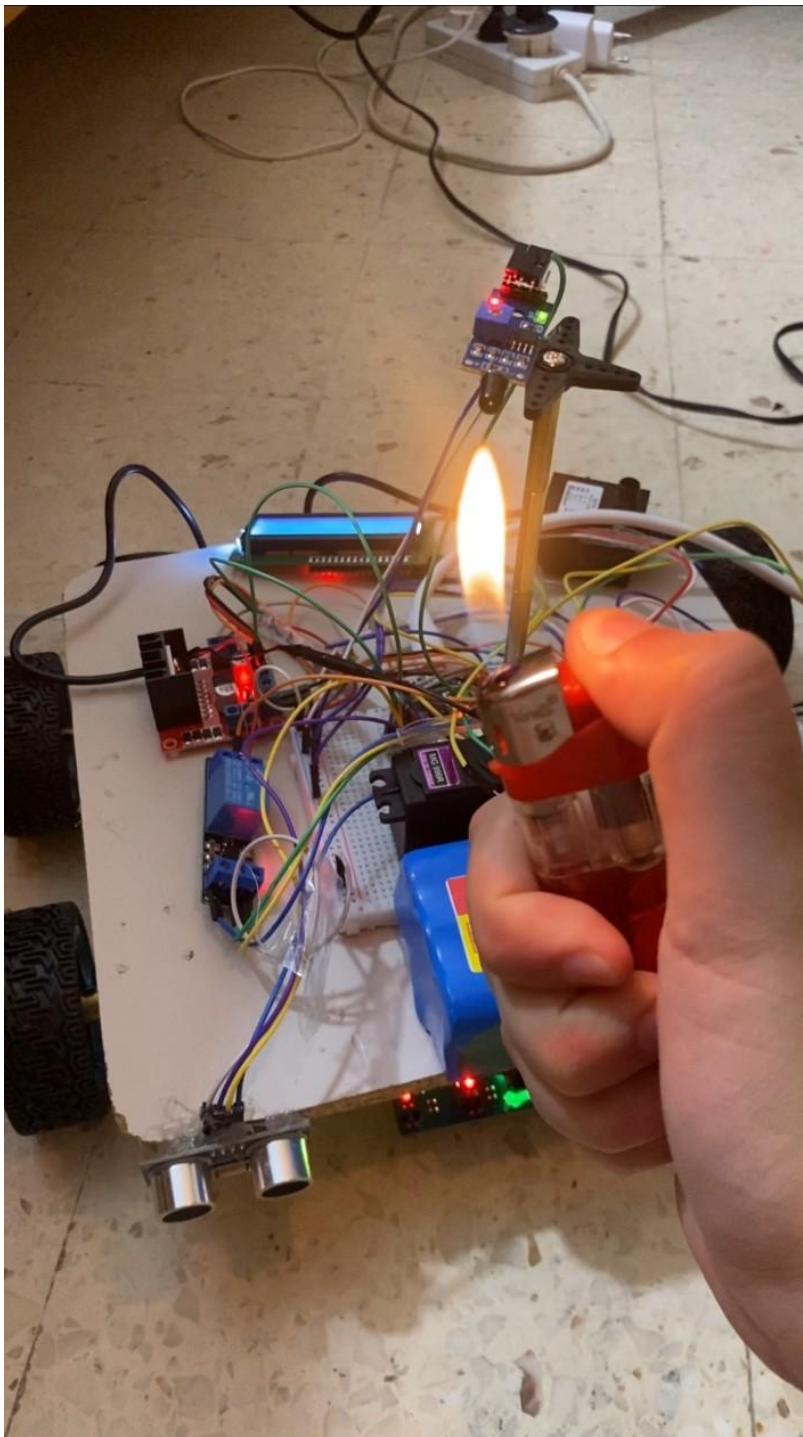


Figure 3. Flame-sensor testing with a lighter while monitoring the response of the integrated prototype.

4.2 DHT11 Temperature and Humidity Sensor

The DHT11 was selected because one low-cost module provides both temperature and relative-humidity measurements through a single digital data line. Its accuracy and response time are limited, but it is suitable for demonstrating environmental monitoring and live LCD visualization.

4.3 YF-S201 Water-Flow Sensor

The YF-S201 was selected to measure liquid flow during pump operation. Water passing through the sensor rotates an internal rotor and produces digital pulses. The ESP32 counts the pulses using an interrupt on GPIO4. The firmware converts pulse frequency to flow rate and integrates the flow to update the amount of water remaining in the tank.

The LCD does not display the flow rate directly. Flow-rate values are available in the CSV serial output, while the LCD shows practical tank information. At startup, the water display begins at W:300. When the stored tank value becomes low, the display shows a refill warning and the remaining amount or percentage.

4.4 Line-Tracking Sensor

A five-channel infrared line-tracking board was used for autonomous path detection. Three outputs were used in the final arrangement: OUT2 for the left reading, OUT3 for the center reading, and OUT4 for the right reading. Three readings were sufficient to implement forward movement and left/right correction while preserving ESP32 pins for the other subsystems.

4.5 HC-SR04 Ultrasonic Sensor

The HC-SR04 was selected for obstacle detection because it provides direct time-of-flight distance measurement and is inexpensive. A trigger pulse is sent by the ESP32, and the returned echo pulse width is converted to distance. The final autonomous logic uses a short stopping threshold of approximately 5 cm because the available demonstration



track is small.

Figure 4. Front-mounted ultrasonic and line-tracking sensors used for obstacle and path detection.

4.6 Pressure Sensor and Force-Sensing Limitation

A pressure-sensing module was considered during development but did not produce reliable readings and was removed from the working prototype. A 5 kg load cell and HX711 interface were also purchased for the force modality. They were not integrated after the main sensing, movement, display, communication, pump, and servo pins had been assigned. This means that the final working prototype did not provide an active force measurement. The load cell and HX711 remain listed in the Bill of Materials as components considered during the design process.

5. Hardware Design and Schematics

5.1 Mechanical Chassis

The chassis was manufactured by the team instead of using a commercial robot frame. A board approximately 25 cm by 25 cm was prepared as the base. Holes and cut-outs were made manually for the bearing and wheel/motor assembly. The handmade approach reduced cost and allowed the team to adapt the placement of sensors, battery, pump, water tubing, LCD, motor driver, and control electronics.



Figure 5. Manual preparation of the chassis material during mechanical construction.

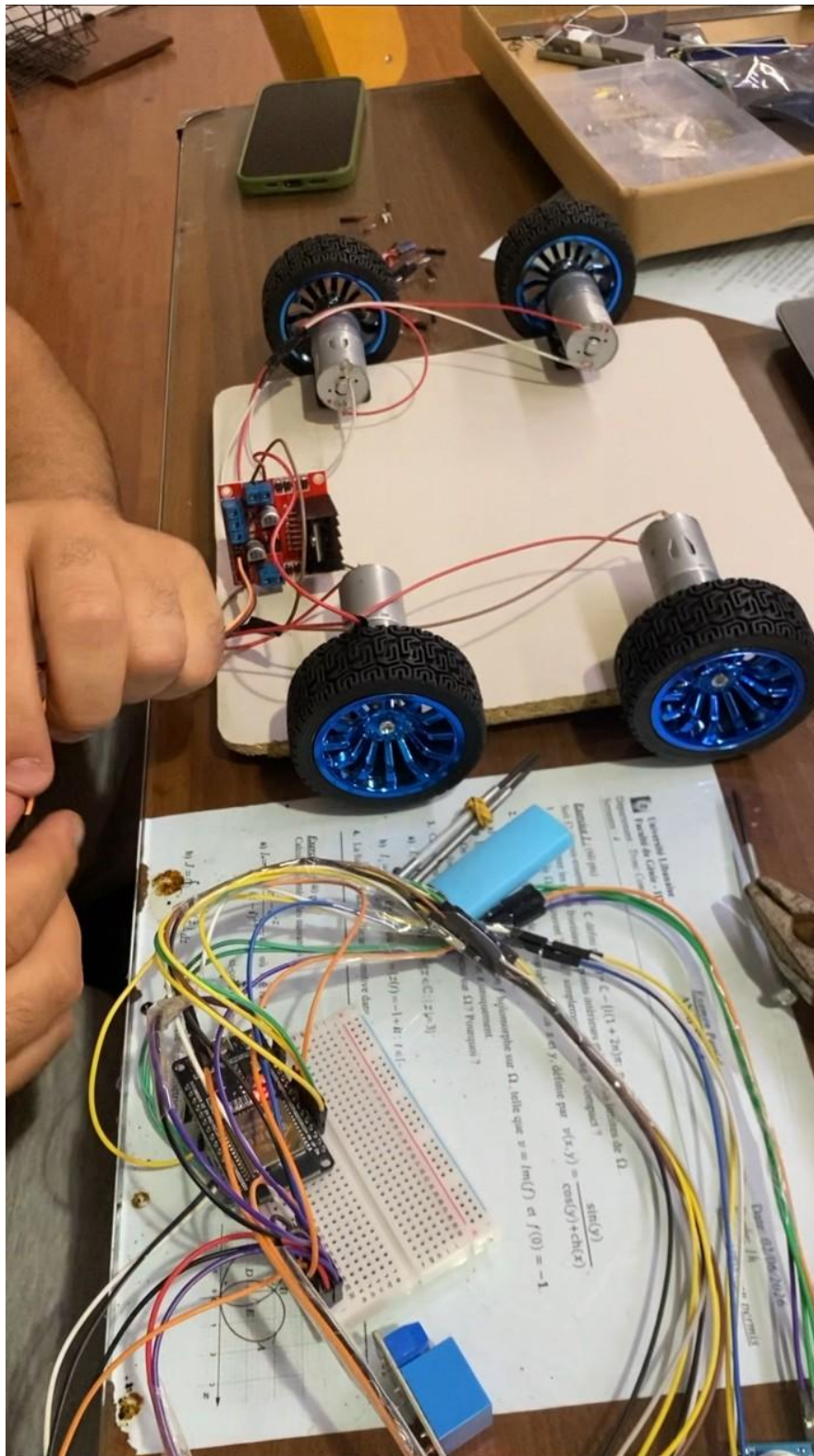


Figure 6. Early motor and wheel assembly on the handmade chassis.

5.2 Electrical Connections

Table 2. Final ESP32 pin assignments used in the autonomous firmware.

Component / signal	ESP32 pin	Notes
LCD SDA	GPIO22	I2C data
LCD SCL	GPIO21	I2C clock
Flame sensor AO	GPIO34	Analog input
DHT11 DATA	GPIO33	Digital sensor input
Flow sensor signal	GPIO4	Interrupt pulse input
Line OUT2 / left	GPIO35	Digital input
Line OUT3 / center	GPIO32	Digital input
Line OUT4 / right	GPIO13	Digital input
Ultrasonic TRIG	GPIO5	Digital output
Ultrasonic ECHO	GPIO18	Input through voltage divider
Servo signal	GPIO23	50 Hz servo control
Relay input	GPIO16	Pump switching
L298N ENA + ENB	GPIO17	Shared motor enable
L298N IN1	GPIO14	Motor direction
L298N IN2	GPIO26	Motor direction
L298N IN3	GPIO27	Motor direction
L298N IN4	GPIO25	Motor direction

5.3 Power Distribution

The 12 V lithium-ion battery supplies the main system. The motor driver and pump use the 12 V source, while the XL4015 buck converter reduces the voltage to approximately 5 V for the servo and suitable logic modules. The buck converter display stopped illuminating during development; therefore, a multimeter was used to adjust and verify the output voltage.

The MG996R servo was powered from the regulated supply rather than from the ESP32 because it requires a higher current. All grounds were connected together to establish a common electrical reference.

5.4 Protection and Hardware Reliability

The HC-SR04 echo output can reach 5 V, so a resistor divider was used before the ESP32 input. The first divider arrangement did not operate correctly and required rewiring. The relay module was damaged during pump testing and was replaced. The original battery charger was also lost, so a 12 V, 1 A replacement charger was purchased. The replacement charger was functional but charged the battery more slowly.

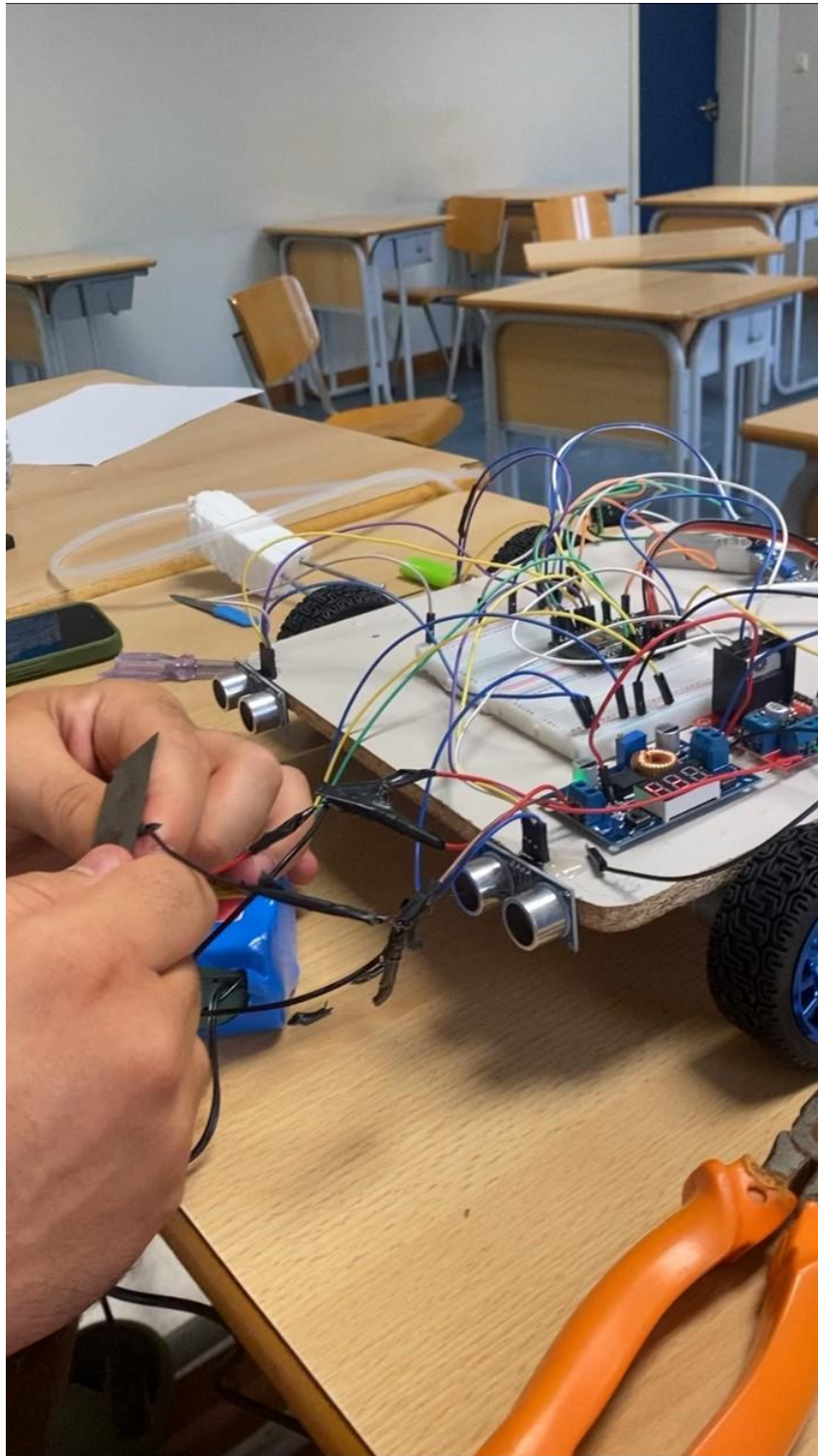


Figure 7. Wiring and integration work during a group assembly session.

6. Software Design and Firmware Architecture

The final firmware is written in Arduino C/C++ for the ESP32. It contains modular functions for sensor acquisition, line following, ultrasonic measurement, fire response, flow calculation, servo scanning, LCD updates, BLE command handling, motor control, and CSV logging.

Table 3. Main firmware functions.

Firmware function	Purpose
readSensors()	Reads flame, DHT11, and line-tracking inputs.
updateUltrasonic()	Measures distance and updates the obstacle state.
followLine()	Selects forward, left, right, or stop based on the three line inputs.
updateFireResponse()	Activates the firefighting response after confirmed flame detection.
updatePumpFlow()	Counts YF-S201 pulses, calculates flow, and updates remaining tank volume.
updateServoScan()	Moves the nozzle while the pump is off.
handleCommand()	Processes BLE commands for movement, pump, servo, and test functions.
updateLCD()	Displays normal measurements, fire warning, obstacle warning, and refill warning.
logData()	Outputs timestamped CSV data through the Serial Monitor.

6.1 Autonomous Navigation

Autonomous mode is enabled by default. When there is no active fire, obstacle, pump operation, or temporary phone-control command, the line-following function evaluates the left, center, and right line states. The center sensor commands forward movement, while the left and right sensors request directional correction. If no valid line pattern is available, the motors stop.

6.2 Obstacle Detection

The ultrasonic sensor is sampled repeatedly. The distance is obtained from the echo pulse using the speed of sound. A confirmation count and release distance can be used to reduce rapid switching near the threshold. When the distance is approximately 5 cm or less, the motors are stopped and the LCD displays OBSTACLE DETECTED.

6.3 Firefighting Response

The analog flame reading is compared with a software threshold. Consecutive qualifying readings can be required to reduce false activation. When fire is confirmed, the robot stops, the nozzle scan stops, the relay switches the pump on, and the LCD displays the fire state. Movement commands are blocked while the pump is active.

6.4 Water-Flow and Tank Calculation

The YF-S201 pulse counter is read once per second. Using the commonly applied calibration factor, pulse frequency divided by 7.5 gives flow in litres per minute. The result is converted to millilitres per second and integrated to calculate total water used. The stored tank capacity begins at 300 mL, and the remaining value is calculated as initial capacity minus accumulated usage.

6.5 LCD User Interface

The display prioritizes information that is useful during operation. The normal screen shows temperature, humidity, and the remaining tank value, beginning at W:300. The LCD does not show the numeric flow rate. When the robot detects fire or an obstacle, the normal screen is replaced by a warning. When the remaining water becomes low, a tank-refill message and the remaining amount or percentage are displayed.

6.6 BLE Control

BLE provides an optional direct-control and testing interface. Phone commands can move the robot, stop it, operate the pump, and enable or stop the servo. Autonomous line following resumes after the temporary direct-control period, unless autonomous operation is explicitly disabled.

7. Calibration Procedures and Results

Calibration evidence note

The course brief requested at least three known reference points for every sensor. The team retained useful functional measurements, but a complete three-point numerical calibration record was not preserved for every modality. The available evidence and limitations are reported below without inventing missing data.

7.1 Flame Sensor

The digital output was first adjusted using the onboard potentiometer, but the response remained insufficient. The analog output was then monitored in the Serial Monitor. With no fire, the ADC reading was approximately 4095. Before improving the sensitivity, a very close flame reduced the reading to approximately 3800. After adjustment of the infrared sensing element/module orientation and sensitivity, the sensor could detect a flame at approximately 30 cm or slightly less.

Table 4. Retained flame-sensor calibration observations.

Test condition	Observed result
No flame	ADC reading approximately 4095
Very close flame before final adjustment	ADC reading approximately 3800
After adjustment	Reliable response at approximately 30 cm or slightly less

The final code used a software threshold selected from the observed analog range. Because the exact final threshold was changed during troubleshooting, the report treats the threshold as a calibrated firmware constant rather than claiming a single laboratory-grade value.

7.2 Pump and Flow Sensor

The pump was tested by collecting a known water volume and measuring the discharge time. A volume of 200 mL was delivered in 8 s, corresponding to an observed pump discharge of 25 mL/s under that test condition.

Table 5. Pump discharge test.

Measured volume	Measured time	Calculated discharge
200 mL	8 s	25 mL/s

The YF-S201 firmware calculation used the factor 7.5: flow in L/min equals pulse frequency in Hz divided by 7.5. This factor provides an initial conversion, while the collected-volume test was used as a practical performance check for the pump and water system.

7.3 Ultrasonic Sensor

The final demonstration used a short obstacle threshold of approximately 5 cm. In direct testing, the robot stopped with an obstacle less than 5 cm away. While following the line, the observed stopping position was approximately 5 to 10 cm because of robot motion, sensor update delay, target angle, and mechanical stopping distance.

Table 6. Ultrasonic obstacle-response observations.

Situation	Observed stopping behavior
Stationary/direct obstacle test	Stopped at less than approximately 5 cm
Moving on line track	Stopped at approximately 5-10 cm
Display response	OBSTACLE DETECTED

7.4 Line Sensor

The line sensor was tested over contrasting light and dark surfaces. OUT2, OUT3, and OUT4 were selected because they produced the most useful left, center, and right information. The module potentiometers were adjusted to improve detection. Stability remained sensitive to lighting, sensor height, and track contrast.

7.5 DHT11

The DHT11 was functionally checked by comparing the displayed measurements with expected indoor conditions. A separate calibrated reference instrument and three-point environmental chamber test were not available. Consequently, the DHT11 results demonstrate acquisition and visualization rather than high-accuracy calibration.

8. Experimental Results and Data Analysis

The final demonstration video presented the robot, its components, direct phone control, obstacle response, line following, fire response, water spraying, refill warning, and normal LCD measurements.

8.1 Demonstrated Behaviors

Table 7. Final demonstration results.

Demonstration step	Observed result
Robot and component overview	The chassis, sensors, LCD, pump, tubing, servo, battery, and electronics were shown.
BLE control	The robot responded to direct movement commands from the phone.
Obstacle test	The robot stopped close to the object and the LCD displayed an obstacle warning.
Line-track test	The robot moved along the small track and stopped before an obstacle.
Firefighting test	The pump sprayed water toward the flame while

Demonstration step	Observed result
	the robot and servo were stopped.
Normal LCD screen	Temperature, humidity, and tank level were displayed.
Low-water LCD screen	A tank refill warning and remaining amount or percentage were displayed.

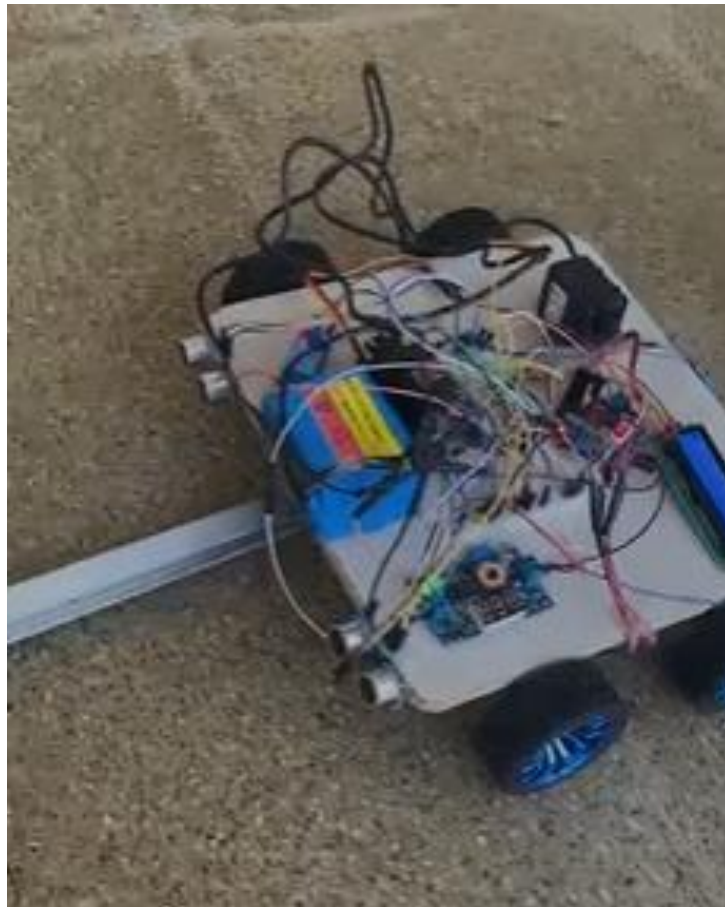




Figure 8. Final firefighting demonstration with the robot directing water toward a flame.

8.2 Data Logging

The firmware produces timestamped CSV output containing temperature, humidity, raw flame reading, fire state, pump state, flow rate, calculated remaining water, ultrasonic distance, obstacle state, line-sensor states, and the latest command. This format supports direct import into spreadsheet or analysis software. The repository contains a demo log file, although the main focus of the final presentation was real-time operation and LCD visualization.

8.3 Interpretation

The demonstration confirmed that the integrated system could execute the key response chain: sense the environment, make a decision, stop unsafe movement, activate the firefighting actuator, and communicate the current state. The system was most reliable on a high-contrast line, with a close and solid obstacle target, and with the flame sensor aimed directly toward the source.

9. Measurement Uncertainty and Error Analysis

Table 8. Main uncertainty sources and mitigation.

Subsystem	Main uncertainty sources	Effect / mitigation
Flame sensor	Ambient light, flame distance, alignment, sensitivity setting	Analog reading and software threshold improved flexibility; orientation remained important.
DHT11	Low sensor accuracy, slow response, indoor air variation	Used for basic monitoring rather than precision measurement.
YF-S201 / pump	Leaks, rotor tolerance, air bubbles, tubing, pressure, battery voltage	Leaks were fixed; collected-volume test provided a practical check.
Ultrasonic	Object angle, target material, very short range, update delay, robot momentum	Short threshold and automatic stop were used; observed stop distance varied.
Line sensor	Lighting, surface reflectivity, height, alignment, potentiometer setting	Three useful channels and a high-contrast track were used.
Tank calculation	Flow conversion factor and integration error	Remaining water is an operational estimate derived from measured pulse flow.

Repeatability was assessed mainly through repeated functional trials rather than a complete statistical study. The robot repeatedly demonstrated the intended state changes, but exact distances, flame range, and water consumption varied between trials. Future work should record repeated numerical measurements at fixed reference points and calculate mean error, standard deviation, and confidence intervals.

10. Engineering Constraints Discussion

10.1 Cost and Availability

Low-cost modules were selected, and the chassis was made manually. Component availability affected the design. Missing or failed parts had to be repurchased, including the relay module and battery charger.

10.2 Reliability and Robustness

The project experienced unstable flame sensitivity, line-sensor variation, water leakage, an unsuccessful pressure sensor, a failed relay, and a non-illuminating buck-converter display. Each issue required a practical workaround, replacement, or redesign.

10.3 Electrical Safety

The system remained at or below 12 V DC. The pump was switched through a relay, the motors through the L298N, and the servo through a regulated 5 V supply. The ultrasonic echo required voltage reduction before the ESP32 input. The battery was charged with a dedicated 12 V charger.

10.4 Power Consumption

The pump, four motors, and MG996R servo are the dominant loads. Their current demand required a separate battery and buck converter rather than relying on the ESP32 supply. The replacement 12 V, 1 A charger increased charging time.

10.5 Maintainability

Separate module test sketches were retained for the servo, flame sensor, ultrasonic sensor, LCD, line sensor, and flow sensor. The GitHub repository separates code, documentation, data, video, and report files, making the design easier to inspect and modify.

10.6 Scalability and Future Improvements

- Integrate the load cell and HX711 using an I/O expander, a second microcontroller, or a redesigned pin map.
- Use multiple flame sensors or a thermal sensor for wider and more reliable fire localization.
- Increase the ultrasonic stopping distance for higher robot speed.
- Apply filtering or closed-loop control to improve line following.
- Add sealed fittings and a more robust tank/tubing arrangement.
- Use a stronger and protected power-distribution board.
- Add autonomous obstacle bypass rather than stopping only.
- Record complete three-point calibration datasets and numerical error statistics for every sensor.

11. Conclusion and Future Work

The project produced a working ESP32-based autonomous firefighting robot that combines sensing, navigation, control, visualization, wireless operation, and a water-delivery actuator. The robot can follow a simple line, stop near an obstacle, detect a flame, stop its movement and nozzle scan, activate a pump, monitor water flow, update the tank state, display environmental information, and log CSV data.

The development process demonstrated the practical challenges of embedded-system integration. The team moved from digital to analog flame sensing, fixed water leakage, reworked the ultrasonic divider, replaced a damaged relay, verified a buck converter with a multimeter, tested the servo separately, and manufactured a custom chassis. These iterations were central to achieving a functional prototype.

The principal limitation is that the force modality was not integrated into the final working robot. Future work should resolve this requirement, improve calibration evidence, increase navigation reliability, and strengthen the mechanical and power systems.

12. References

- [1] Lebanese University, Faculty of Engineering III. Sensors & Instrumentation Final Project Proposal, Academic Year 2025-2026.
- [2] Espressif Systems. Arduino-ESP32 documentation and ESP32 technical documentation.
- [3] Adafruit Industries. DHT Sensor Library documentation.
- [4] ESP32Servo Library documentation.
- [5] LiquidCrystal_I2C Arduino library documentation.
- [6] YF-S201 water-flow sensor module technical documentation.
- [7] HC-SR04 ultrasonic sensor module technical documentation.
- [8] L298N dual H-bridge motor-driver module technical documentation.
- [9] Project GitHub repository: <https://github.com/nourchamas09-crypto/Sensors-Instrumentation--Final-Project->
- [10] AI tools were used as development assistants for debugging, code organization, and documentation drafting. The submitted design, code, and report must be reviewed, tested, understood, and explainable by the team.



Appendix A. Demonstration Sequence

1. Show the complete robot and identify the main sensing and actuation components.
2. Demonstrate phone-based movement control as an optional testing interface.
3. Place an object close to the robot and show the automatic stop and LCD obstacle warning.
4. Demonstrate autonomous movement on the line track.
5. Present the flame-detection and water-spraying sequence.
6. Show the LCD normal screen with temperature, humidity, and W:300 tank information.
7. Show the tank-refill warning and remaining amount or percentage.

Appendix B. Repository Organization

The project repository is organized as follows:

- README.md
- docs/ - BOM, pinout, architecture, wiring, troubleshooting, development notes, and circuit diagram
- code/ - autonomous firmware and separate module test sketches
- data/ - CSV demonstration log
- video/ - demonstration video link
- report/ - final report and individual contribution statements

Appendix C. Team Roles Summary

Table 9. Summary of team contributions. Detailed one-page statements are stored separately in the repository.

Team member	Main documented contributions
Nour Chamas (6890)	Component purchasing and replacement, attendance at all work sessions, assembly/wiring/testing support, troubleshooting, final enclosure, filming and editing the presentation video, GitHub organization and documentation.
Hassan Awala (6602)	Sensor-selection research, calculations, L298N motor subsystem and wheel installation, ultrasonic/flow calculations, motor testing, and technical documentation.
Hawraa Diab (6861)	Manual wiring, hardware assembly, sensor integration, code testing and modification support, troubleshooting, system integration, repository organization, and final demonstration preparation.